ADEMP protocol: 05-nof1-design-sensitivity

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This is the working protocol for the exploratory simulation study behind `analysis/report/05-nof1-design-sensitivity/`. It replaces the protocol of 2026-05-07, whose grids matched none of the drivers. The study is exploratory. No analysis plan was fixed in advance, and the findings need confirmation in an independent study. The ADEMP structure follows Morris, White and Crowther (2019).

Aims

1. Describe the power of the Hendrickson et al. (2020) Hybrid design for the treatment main effect as one factor at a time departs from a baseline.
2. Find the regimes in which the Hybrid has more or less power than an alternating design, a two-period crossover and a parallel group on a common estimand.
3. Report Type I error, bias and standard-error calibration for every design.

Data-generating mechanisms

- Outcome $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$. Modified Gompertz means (displacement 1, rate 0.3 per week), with asymptotes 0.5 (TV), 1 (PB) and a per-design value for BR. Component SDs 0.5, 0.5, 1. Baseline score mean 8, SD 1.5. Biomarker mean 120, SD 15.
- AR(1) correlation $\rho = 0.3$ per week within components, 0.1 between components at the same visit, 0.05 times the AR(1) factor at other visits. $c_{bm} = 0.3$.
- Carryover on the BR mean, exponential, half-life 3 days (3/7 week), with the original (compounding) rule of `docs/53`.
- Participant-level heterogeneity of the drug effect, SD τ (baseline 0).
- $N = 35$ in total, split over the paths by the engine rule.
- Designs, all with eight follow-up visits: Hybrid (published, weeks 4, 8, 9, 10, 11, 12, 16, 20, four paths), Alt (alternating, two paths, same weeks), XO (two periods, weeks 10 and 20), RCT (two arms, same weeks, expectancy 0.5), RCT-OL (expectancy 1) and Compact2 (the two-path weekly schedule of the 2026-05 runs, S1 only). Expectancy is 0.5 at blinded visits and 1 at open-label visits.
- **Common estimand.** The drug-response asymptote is set per design so that Δ_{on}, the mean drug response over the on-drug follow-up visits, equals -0.5 outcome units (S1 sweeps it). Every artifact stores `delta_on`.

- A cell whose covariance matrix is not positive definite is dropped, never repaired (`stop_on_nonpd = TRUE`). The artifact stores the dropped cells with the smallest eigenvalue.

Estimands

The treatment main effect β_D: the coefficient of the exposure indicator for the within-subject designs, and the arm coefficient of the ANCOVA for the parallel group. The target of bias and coverage is its large-sample value (`ETbeta`, the coefficient of a fit to all simulated participants of the cell). The difference from Δ_{on} is reported as well.

Methods

- Within-subject designs: `nlme::lme`, random intercept per participant, `corCAR1(~t | ptID)` residuals, fixed part `Sx ~ bm + t + Dbc` plus `De` where expectancy varies.
- Parallel groups: follow-up mean regressed on arm and baseline score by least squares (`parallel_analysis = 'ancova'`). The mixed-model analysis of 2026-05 is kept as `'legacy'` and checked in script 14.
- Assumed half-life of the exposure indicator: 0.2 week in all sweeps except S2 (0, 0.2), S3 (0, 0.2, 3/7), S11 (0, 0.2, 3/7) and S12 (0, 0.2).

Performance measures

Rejection rate at 0.05, bias against `ETbeta`, bias against Δ_{on}, empirical SD, mean model SE, κ, coverage of the 95% interval, share of failed fits, each with its Monte Carlo standard error. A Type I error is nominal when it lies within two MCSE of 0.05 (MCSE at 0.05).

Sweeps

Sweep	Axis	Grid	Designs	Reps
S1	Δ_{on}	0, -0.25, -0.5, -0.75, -1, -1.5, -2	Hybrid, Compact2, Alt, XO, RCT, RCT-OL	1000

Sweep	Axis	Grid	Designs	Reps
S2	DGP half-life (days) by assumed half-life	0, 0.5, 1, 2, 3, 5, 7, 10, 14; 0, 0.2 wk	Hybrid, Alt, XO, RCT	500
S3	decay form by assumed half-life	exponential, linear, Weibull 0.8, Weibull 1.5, power; 0, 0.2, 3/7 wk	Hybrid, Alt, RCT	500
S4	τ; baseline SD	0, 0.25, 0.5, 0.75, 1; 0.5, 1, 1.5, 2, 3	Hybrid, Alt, RCT	500
S5	multiplier on the three component SDs	0.5, 0.75, 1, 1.25, 1.5	Hybrid, Alt, RCT	500
S6	periods in 16 weekly visits	2, 4, 8, 16 (RCT: 16 visits)	Alt, RCT	500
S7	total N	16, 24, 35, 60, 80, 120	Hybrid, Alt, RCT	500
S8	AR(1) ρ	0, 0.3, 0.5, 0.7, 0.9	Hybrid, Alt, RCT	500
S9	period (days) by half-life (days)	3, 5, 7, 10, 14; 0.5, 3, 7	Alt	500
S10	c_{bm} by encoding by half-life	0, 0.2, 0.4, 0.6; mvn, mean_moderation; 0, 3 days	Hybrid, Alt, RCT	500
S11	DGP carryover by DGP AR(1) by effect, assumed half-life	off/3 d; off/0.7; null/-0.5; 0, 0.2, 3/7 wk	Hybrid, Alt, RCT	500

Sweep	Axis	Grid	Designs	Reps
S12	null, four configurations	BL SD 1.5/1.5/0.75/2, BR SD 1/1/1.5/0.5, half-life 0/3/3/3 d; assumed 0, 0.2 wk	Hybrid, Alt, XO, RCT	2000
S12b	refit without corCAR1	null, baseline configuration	Alt, XO	2000
S14	parallel-group check	rho 0, 0.3, 0.7; effect 0, -0.5; legacy, ancova	RCT, RCT-OL	1000

Deliverables

- Drivers at `analysis/scripts/nof1-design-sensitivity/`. Every driver asserts that the parameter values stored in the artifact equal its grid and that run plus dropped cells equal the grid size. A failed assertion saves the artifact as `*-UNCHECKED.rds`.
- Artifacts `sensitivity-NN-name.rds` with the run metadata (`run_meta`: schema, replicates, seed, engine version, revision).
- `cell-summaries.csv` from `16-summarize-all.R`.
- Figures from `plot-all.R`.

Seeds

Each driver calls `set.seed(42 + id)` with the sweep number as `id`. Cells run in parallel with `furrr` and `seed = TRUE`. There is no separate seed file.

Deviations log

- 2026-10-28. Protocol rewritten after the referee report of 2026-10-28 (`analysis/report/05-nof1-design-sensitivity/ referee-report-2026-10-28.md`). Changes: published Hybrid schedule; common estimand; ANCOVA for the parallel group; analysis half-life crossed with the DGP half-life in S2, S3, S11, S12; feasibility check with dropped cells; S4 now varies τ (added to the DGP) and, separately, the baseline SD; S5 scales all component SDs; S6 fixes the trial length; S9 passes the whole grid; S12b

and S14 added; baseline effect changed from a response asymptote of 2 to $\Delta_{\text{on}} = -0.5$.
- The earlier references to `01-base-seed.txt`, `02-deviations-log.md`, `02-grid-summary.rds` and `docs/sensitivity-sweep-plan-2026-04-17.md` are withdrawn. None of these files exists.